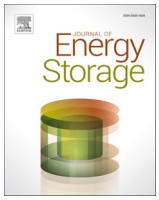
Journal of Energy Storage 164 (2026) 121839



Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est



Research Papers

A prospective life cycle assessment of different battery technologies for mobility, including a second stationary application

Luis Alfredo Carmona Arrieta [a,b,*], Paula Sanz Camacho [b], Juan M. Perez [b], Christian Bauer [c], David Parra [b,*]

[a] Universidad Politécnica de Madrid, Escuela Técnica Superior de Ingenieros de Minas y Energía, Calle Ríos Rosas, 21, 28003, Madrid, Spain
[b] Iberian Centre for Research in Energy Storage (CIIAE), Cáceres 10003, Spain
[c] Laboratory for Energy Systems Analysis, PSI Centers for Energy and Environmental Sciences and Nuclear Engineering and Sciences, Paul Scherrer Institute, Villigen, Switzerland

ARTICLE INFO

Keywords:
Life cycle assessment (LCA)
Prospective life cycle assessment (*p*-LCA)
Lithium-ion batteries (LIB)
Sodium-ion batteries (SIB)
Solid-state batteries (SSB)

ABSTRACT

The versatile nature of batteries makes them an essential component of both stationary and mobility applications, despite their different requirements, throughout the energy transition. Batteries used in electric vehicles (EVs) can further be used in stationary applications, such as for community PV self-consumption. This second application can indeed reduce environmental burdens by extending the usage phase. Here, we analyse different battery technologies, namely, lithium-ion batteries (LIB), sodium-ion batteries (SIB), and solid-state batteries (SSB), and compare their current and potential future environmental impacts using (prospective) Life Cycle Assessment (LCA) methodology, following a cradle-to-grave approach depending on the type of application, namely mobility, stationary, or combined. Our results for combined applications indicate climate impacts around 88, 95, and 96 kgCO2-eq/kWh of nominal battery storage capacity for LIB (NMC622), SIB (NaNMMT), and SSB (NMC900), respectively. Furthermore, a potential greenhouse gas emission reduction in the range of 66–72% can be achieved for the same chemistries based on the results of a prospective LCA (*p*-LCA). Additionally, a sensitivity analysis is performed to determine the impact of both applications, namely mobility and stationary, on emissions attribution. Finally, other impact categories such as acidification, material resources, and water use are further considered to perform a more inclusive environmental analysis. Thus, we highlight the need to balance critical and abundant raw material incorporation to enhance both the technical performance and long-term environmental sustainability of various battery technologies.

1. Introduction

After the Paris Agreement, the European Commission (EC) presented a strategic long-term plan to become the world's first carbon-neutral continent by 2050 [1,2]. To reach this goal, renewable energy technologies should increase their participation in the energy supply market from 14% in 2017 to 86% (i.e., a total generation of 47,000 TWh per annum) by 2050 [3]. However, key renewable energy technologies such as solar and wind cannot supply electricity on demand since they are highly dependent on weather conditions. Therefore, energy storage is needed to increase their value by converting intermittent technologies into flexible supply technologies [4,5]. Among different energy storage technologies, battery energy storage systems are becoming essential to support renewable energy technologies by performing short-term discharges (seconds to minutes) and mid-term discharges (several hours) in key sectors such as residential and industry, but also to decarbonise the transport sector [6].

Consequently, the European Union (EU) has worked on specific strategies concerning battery regulations to conceive standard, open and clear information about their production, transport and distribution, use, reuse, recycling, and waste management, as in the Regulation (EU) 2023/1542 concerning batteries and waste batteries [7]. For instance, by 2036, new batteries with cobalt, lead, lithium, or nickel as active materials should contain at least 26%, 85%, 12% and 15% of recycled materials, respectively [7]. Besides, a digital battery passport will be required for every industrial or electric vehicle (EV) battery with a capacity higher than 2 kWh by 2027, according to the Art. 77 of the aforementioned regulation [7]. This European regulation aims to

* Corresponding authors.
E-mail addresses: luis.carmona@ciiae.org (L.A.C. Arrieta), david.parra@ciiae.org (D. Parra).

https://doi.org/10.1016/j.est.2026.121839
Received 10 December 2025; Received in revised form 28 February 2026; Accepted 23 March 2026
Available online 15 April 2026
2352-152X/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

provide guidelines on how to implement and track battery information digitally. For example, information concerning the battery manufacturer, location, chemistry, hazardous substances, critical raw materials, and carbon footprint should be provided for every battery in the EU market [7].

The carbon footprint declaration should be calculated throughout the battery life cycle, as expressed by the Art. 7 of the Regulation (EU) 2023/1542 [7]. Standardised by ISO 14040 (2006), ISO 14044 (2006) and ISO 14072 (2024), the Life Cycle Assessment (LCA) methodology is widely used to evaluate the environmental emissions of a product [8–10]. Step by step, the LCA methodology guides practitioners to define the goal and scope, collect the process inventory, calculate the environmental impact, and finally interpret the results obtained [11]. Thus, the carbon footprint of a battery can be measured in terms of impact on climate change, but also other impact categories, such as acidification, resource use, water use, and human toxicity, which are further quantified to provide a holistic view of environmental impacts [12].

Previous studies of LCA of batteries have been mainly performed for Lithium-Ion batteries (LIB), which is currently the most prominent battery technology in the market [13]. LIB can provide high energy and power density, efficiency, charge-discharge lifecycles, and also, relatively low self-discharge, which make them very suitable for electronic devices, EVs, and also in stationary storage systems [14]. For mobility applications, the Lithium-Nickel-Manganese-Cobalt (NMC) chemistry dominates the market by accounting for 52% of the EVs (2023), due to its good performance [6,15,16]. Additionally, when compared to other emerging technologies, mainly Sodium-Ion batteries (SIB) or Solid-State batteries (SSB), LIB still show higher performance [17]. However, there are some concerns about geopolitical disruptions that put LIB into a vulnerable supply chain risk, especially the geographical limitations associated with critical raw materials, namely lithium, nickel, cobalt, and copper [18].

From a LCA perspective, LIB has been widely studied for mobility applications [17,19,20]. For instance, some authors limited their analysis to the emissions expected from the LIB production process [20], while others [19] extended their analysis to the battery use in a plug-in hybrid EV [21]. Moreover, Degen et al. (2024) took industrial manufacturing data to estimate the environmental emissions from LIB and SSB, in particular for high-power and high-energy batteries that are used currently in the EV market, depending on the needs [17]. From an application perspective, Safari (2025) and Chen (2025) focused on EV batteries, considering the potential supply chain disruption created by the geo-limitations of the LIB raw materials [22,23].

Additionally, the LCA methodology has also been applied to batteries in stationary applications [24–26]. Rossi et al. (2020) analysed a PV home system coupled with a battery to estimate the environmental feasibility for several European countries, including both LIB and SIB [24]. A similar analysis was done by Jasper et al. (2022), who, based on primary data from a home storage energy system, estimated the environmental emissions of both LIB and SIB [26]. Additionally, Schmidt et al. (2019) analysed the battery environmental and economic burdens that are expected from stationary applications associated with their geography [27]. Importanly, other authors like Peters et al. (2021) further included the recycling battery process to highlight not only the process efficiency, but also the low environmental emissions of an advanced hydrometallurgical treatment, in addition to the production and use phase [25].

Interestingly, due to the criticality and the geographical scarcity of the LIB raw materials, the reuse of batteries is a promising alternative to maximise the useful life of the battery. An EV battery can be further used for stationary applications, which are typically less demanding in terms of power and cooling conditions. This is particularly the case for NMC111 and NMC811 chemistries [28,29]. A second life approach was used to compare LFP (LIB) and NaNMMT (SIB) chemistries in a previous study [30]. The authors considered the battery production, first use in EV, a second use in communication base stations, recycling and disposal, and concluded that batteries could reduce their environmental emissions during their whole life cycle, based on impact categories different than global warming potential, such as the ozone depletion, resource depletion, and terrestrial acidification, among others [30]. Independently of performing one or two applications by the same battery (LIB, SIB or SSB), the Regulation (EU) 2023/1542 demands that the recycling efficiency shall achieve at least 70% by average weight of lithium-based batteries by 2030, while all recycling shall achieve the following targets for recovery of materials by 2031: 95% for cobalt, copper, lead and nickel, and 80% for lithium [7].

To the best of our knowledge, no study has so far considered the production or the recycling of SSB for a second life application, despite being an emerging technology that is gradually gaining market implementation [31–33]. Moreover, a prospective analysis for LIB, SIB and SSB during their second life application could provide clear pathways about their future implementation and viability [34–36].

Based on this literature review, we identify some relevant research gaps in the existing literature. First, there is still a limited number of LCA of batteries with a second application. Secondly, *p*-LCA has not yet been used to establish future potential emissions for batteries that perform a second application. This prospective analysis could be further enriched when breakdown by battery components to maximise potential reductions. Against this background, in this study, we analyse and compare current and future environmental burdens of LIB, SIB, and SSB for mobility applications. Importantly, we further consider the advantages of performing a second stationary application from an environmental perspective based on the LCA methodology. Furthermore, we take into account a prospective-LCA (*p*-LCA) scope to frame our results in the context of a decarbonised society by 2050. This allows us to estimate different battery technology emissions and trade-offs when having a second application in this future scenario. More importantly, we highlight hotspots to maximise the potential greenhouse gas emission reduction for the different battery technologies in the current as well as future market scenarios.

2. Methodology

In this study, seven different battery cell chemistries are studied under the LCA methodology to determine their environmental burdens throughout their life cycle [9]. The seven battery cell chemistries were chosen both for their availability in the current market and their potential in the future market [17]. Thus, NMC622, NMC811, and LFP chemistries were selected from LIB technology; NaNMMT, NaMVP, and NaPBA chemistries were selected from SIB technology. To compare with another promising emerging technology, NMC900 (SSB) was further selected since it provides both high energy density and low safety concerns associated with liquid-state electrolytes. The main characteristics of the studied battery technologies are shown in Table 1, where charge and discharge efficiency (90%), depth of discharge (80% or 60%, depending on the application), and energy distribution coefficient (0.49), which is the ratio of the battery's electrical energy consumption to the mass of the vehicle, have been considered the same among all the chemistries based on the previous literature [25,30,37]. The use of constat values for these technical parameters can be used to highlight the real benefits that the battery weight, energy density, and charge-discharge lifecycles themselves can provide [25,30,37]. Besides, as per their current development, SIB show lower energy density than LIB and SSB [17,37,38].

2.1. Goal and scope

The goal is to analyse and compare the environmental burdens from the selected battery technologies based on the LCA methodology, depending on the type and number of applications, namely: i) mobility only; ii) both mobility and then stationary as a second application, with key for input datadetails shown in the supplementary information (SI).

Table 1
General characteristics of the various battery technologies considered in this study.

Characteristic	NMC622	NMC811	LFP	NaNMMT	NaMVP	NaPBA	NMC900
Battery pack weight [kg]	160	145	170	190	205	230	125
Energy density [Wh/kg]	197	209	157	157	145	132	265
Charge and discharge efficiency	90%	90%	90%	90%	90%	90%	90%
Depth of Discharge – 1st use	80%	80%	80%	80%	80%	80%	80%
Depth of Discharge – 2nd use	60%	60%	60%	60%	60%	60%	60%

Note: data were collected for NMC811 from [38]; for NMC622, LFP and NMC900 from [17]; and NaNMMT, NaMVP, and NaPBA from [37]. Thus, NMC622 refers to $Li_{1.05}(Ni_{0.6}Mn_{0.2}Co_{0.2})_{0.95}O_2$, NMC811 refers to $Li_{1.05}(Ni_{0.8}Mn_{0.1}Co_{0.1})_{0.95}O_2$, LFP refers to $LiFePO_4$, NaNMMT refers to $Na_{1.1}(Ni_{0.3}Mn_{0.5}Mg_{0.05}Ti_{0.05})O_2$, NaMVP refers to $Na_4MnV(PO_4)_3$, NaPBA refers to $Na_2Fe[Fe(CN)_6]$, and NMC900 refers to $Li_1(Ni_{0.9}Mn_{0.05}Co_{0.05})O_2$.

The environmental emissions considered are generated from production, application either in Mobility or Stationary, recycling, and disposal of the batteries, according to each flow diagram shown in Fig. 1. Thus, the inclusion and exclusion of the second application has environmental impacts associated to each path taken. Here, both cradle-to-gate (C2G) and cradle-to-grave (C2Gr) approaches were considered for the analysis. Moreover, the functional unit to estimate the environmental emissions is 1 kWh of nominal battery storage capacity, which is also commonly used to compare results with previous publications. Furthermore, by having the same operational characteristics among the selected batteries, the same functional unit could be applicable to perform both C2G and C2Gr analyses during the current study.

2.1.1. Production process

The production process of the battery cells encompasses primary raw material extraction, battery cell production, and assembly, placed at the factory gate, ready to be used. This is the baseline for the C2G approach considered in this study [17,25].

2.1.2. Application 1: Mobility use

Here, we assume that the electric motor of an EV has the same consumption for all the various technologies, as follows: 12.2 kWh per 100 km [30]. Besides, since the electricity needed for an EV does not depend on the battery per se, but on the electric motor, only battery losses are considered in this study. Thus, Eqs. (1)–(2) are used to quantify losses due to the battery overcoming its mass (E_m) and due to the battery charging and discharging (i.e. round-trip) efficiency (E_c), respectively [30,39,40]. In this study, and as a consequence of considering consistent technical characteristics, the losses due to charge-discharge efficiency (E_c) are considered the same for the various battery technologies. Additionally, we assume that the nominal battery capacity is reduced to 80% after 200,000 km of use. These assumptions are linked to the warranty given by the car manufacturers and the common use in previous research studies [30,41,42].

Losses due to the battery overcoming its own mass

$$E_m = f \bullet E_p \bullet L \bullet \left(\frac{m}{M}\right) \tag{1}$$

Where:

$E_m = Energy\ loss\ due\ to\ the\ battery\ overcoming\ its\ mass, kWh$

$f = Energy\ distribution\ coefficient\ of\ the\ battery\ control\ system\ during\ EV\ driving, equal\ to\ 0.49$

$E_p = Electric\ energy\ consumption, 12.2\ kWh\ per\ 100\ km$

$L = Total\ mileage\ within\ the\ EV\ use\ phase, 200,000\ km$

$m = Mass\ of\ the\ battery\ system, kg$

$M = Mass\ of\ the\ EV, kg$

Losses due to battery Charge-Discharge Efficiency

$$E_c = E_p \bullet L \bullet (1 - \eta_1) \tag{2}$$

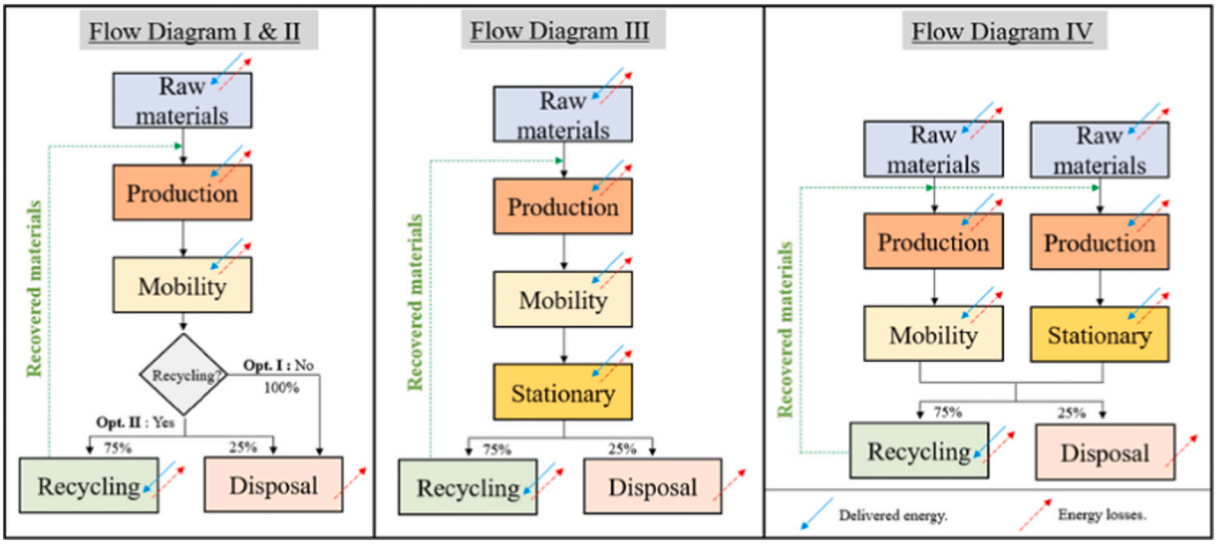


Fig. 1. Process flow diagram of the battery life cycle considered in this study: I) Process flow for battery production used in mobility without recycling; II) Process flow for battery production used in mobility with recycling; III) Process flow to produce a battery to be used both in mobility and stationary with recycling; IV) Process flow to produce two batteries to be used in mobility and stationary with recycling, separately. Here, the recovered materials from the recycling process will be used to mitigate the need of virgin raw material during the production process. Besides, the diagonal arrow (→) represents the delivered energy to each process, and the dashed-arrow (→) represents the energy losses of the process.

Where:

$E_c = Energy\ loss\ due\ to\ the\ battery\ charging\ and\ discharging\ efficiency, kWh$

$E_p = Electric\ energy\ consumption, 12.2\ kWh\ per\ 100\ km$

$L = Total\ mileage\ within\ the\ EV\ use\ phase, 200,000\ km$

$\eta_1 = Charge\ and\ discharge\ efficiency\ of\ the\ battery$

2.1.3. Recycling process

The considerations during the recycling process are based on previous publications [25,43,44], which show the consumption for three different recycling processes: i) the pyrometallurgic recycling process has an efficiency of 93.6% to recover some critical raw materials such as nickel, cobalt and copper, while 98% efficiency to recover steel [43,44]; ii) the hydrometallurgical process could recover 93.6% of some other critical raw materials such as lithium, nickel, cobalt, titanium, steel, aluminium, copper, among others. This wider list of recovered components increases the total efficiency of the recycling process when compared to the pyrometallurgical process [43,44]; iii) the advanced hydrometallurgical process is reported to recover 80% of lithium and titanium, 98% of nickel and cobalt, and 100% of aluminium, copper, graphite and electrolyte. Neither graphite nor the electrolyte are recovered for any of the previous recycling processes [25,45]. The high recovering capacity of the advanced hydrometallurgical recycling process is based on a patent developed by Duesenfeld GmbH (2020), which involves a mechanical pre-treatment before the hydrometallurgical treatment [45]. Here, during the mechanical pre-treatment, solid parts that contain iron, aluminium, and copper are recovered, and during the hydrometallurgical treatment, other metals are recovered, namely lithium, nickel, cobalt, and manganese, among others [45]. Besides, the recycling process is modelled following a product substitution approach, where the recovered materials are assumed to avoid the equivalent extraction of primary raw material needed in the production process. Thus, the recycling process could lead to the reduction of the total climate change emissions of the produced battery. The Duesenfeld GmbH (2020) recycling process is in accordance with the European regulation that aims to reach a recycling efficiency for lithium-based and nickel-cadmium batteries of 70% and 80%, respectively, by 2030 [7,45]. Similarly, the case studies set in this research are in accordance with the Regulation (EU) 2023/1542 that demands a recycling efficiency of at least 70% by average weight of lithium-based batteries by 2030 [7].

2.1.4. Disposal process

While battery recycling is one of the main objectives at the EU level by 2031, when the target is to increase the battery collection rate to 100%, and the critical material recovery to 80–95% [7], we also consider the disposal process, i.e., to quantify the emissions that could be generated by the components when they are not recycled and just disposed in a landfill. Thus, energy, heat, and the emissions to the air previously are omitted in the production phase in order to estimate the emissions of disposing of the battery in a landfill.

2.1.5. Application 2: Stationary use

Eqs. (3)–(4) quantify the losses due to the secondary use of the battery in stationary applications [30,46,47]. This set of equations is based on the charge and discharge cycles that the battery can provide when it is used in stationary applications such as community PV self-consumption, after it has had a first exhausting mobility application [30,48]. The GHG emissions from the stationary application are discarded due to the low emissions associated with PV supply as part of the self-consumption phase. Additionally, we assume that the nominal battery capacity is reduced from 80% to 60% during this application and before it is disposed, reflecting a linear degradation during the second stationary application [30,41,49].

Battery capacity loss with the second stationary application

$$\varphi = \lambda \bullet e^{-(E_a/R \bullet T)} \bullet n^z \tag{3}$$

Where:

$\varphi = Capacity\ loss\ of\ the\ battery, kWh$

$\lambda = Calculation\ constant$

$E_a = Activation\ energy, J \bullet mol^{-1}$

$R = Gas\ constant, J \bullet mol^{-1} \bullet K^{-1}$

$T = Thermodynamic\ temperature, K$

$n = Number\ of\ charge\ and\ discharge\ cycles\ in\ the\ secondary\ application$

$z = Power - law\ coefficcient$

Losses due to battery in second use in stationary applications

$$E_{stat} = \sum_{n=1}^{i} E_f \bullet DOD \bullet (1-\varphi) \bullet \frac{1-\eta_T \bullet \eta_2}{\eta_T \bullet \eta_2} \tag{4}$$

Where:

$i = Indicates\ the\ number\ of\ charges\ and\ discharge\ to\ reach\ the\ battery\ capacity\ to\ 60\%$

$E_{stat} = Energy\ loss\ of\ the\ battery\ in\ second\ stationary\ application, kWh$

$E_f = Initial\ capacity\ of\ the\ battery\ when\ to\ start\ its\ second\ application, kWh$

$DOD = Depth\ of\ discharge\ during\ battery\ use, 80\%$

$\eta_T = Battery\ transfer\ efficiency, 90\%$

$\eta_2 = Charge\ and\ discharge\ efficiency\ during\ secondary\ use, 85\%$

2.1.6. Case studies

To perform the LCA for the battery technologies under a C2Gr approach, four different case studies are considered, shown in Table 2. *Case Study I* includes the battery production (100%) and a single mobility application (100%). After the use phase, 100% of the battery is disposed of without any recycling process. Besides, *Case Study II* includes the battery production (100%) and a single mobility application (100%). Yet, after the use phase, 75% of the battery is recycled, and 25% is disposed of. With these two case studies, the main objective is to illustrate the advantages of recycling raw materials, which will be progressively implemented by the EU from 2026 to 2030 [7].

For Case Study III, the attribution of emissions was done for both mobility and a second stationary application. Since we consider 1 kWh as the functional unit for our research, we could eventually see that the battery degradation affects the nominal storage capacity of the battery. Thus, the allocation of the environmental burdens is given based on the

Table 2
Percentage attribution of the emissions from each process according to the case study.

	Case Study			
Process	I	II	III	IV
Production	100%	100%	100%	200%
Mobility	100%	100%	60%	100%
Stationary	0%	0%	40%	100%
Recycling	0%	75%	75%	150%
Disposal	100%	25%	25%	50%

battery capacity loss that is suffering during its lifetime. Previous publications have quantified the environmental burdens for the use and reuse phase based on the battery capacity loss [30,41,49]. For instance, Guo et al. (2023) and Kang et al. (2025) included the battery capacity loss in their analysis of battery emissions, by accounting for the battery ageing during the use in mobility and reuse in stationary applications, to better describe the battery behaviour and estimate the environmental emissions [28,30]. Here, the initial reduction in nominal capacity is assumed to be equal to 20% for either mobility or stationary. For the remaining 60% of the battery capacity, 66% was attributed to mobility, as this application is more energy demanding application (i.e., a higher nominal power per unit of nominal battery capacity, also it is the main purpose of producing a battery in this study), and the other 33% was attributed to the stationary application. Thus, 60% of the emissions were attributed to mobility, while 40% correspond to the second stationary application, as schematically represented in Fig. 2. Additional to the percentage of attribution, the GHG emissions associated with the combined applications, mobility and stationary, are the result of multiplying the losses calculated previously (see Eqs. (1)–(4) given in kWh) by the emissions associated to the electricity generation of reference, e.g., market group for electricity, medium voltage, Europe without Switzerland, given in kg CO2-eq per kWh.

Therefore, *Case Study III* includes the battery production (100%), and then 60% of the emissions are attributed to the first mobility application, while 40% of the emissions are attributed to the second stationary application. Then, after the total use phase, 75% of the battery produced is recycled, and 25% disposed of. Finally, *Case Study IV* includes the production of two different batteries (200%), one to be used in mobility (100%) and the other one to be used in a stationary application (100%). After each use phase, 75% of each battery (i.e., 150% overall) is recycled, and 25% of each battery (i.e., 50% overall) is disposed. With these two case studies, the main objective is to illustrate the benefits of considering a second-life application, which directly translates to the avoidance of producing new batteries specifically for a stationary application.

2.2. Life cycle inventory (LCI) analysis

The full inventory for all seven batteries used in this study can be found in the SI. The production process inventory is taken from the Premise open-source software online repository [50]. Thus, based on these inventories for the battery cell production, other inventories are prepared accordingly, namely, recycling and disposal. At the same time, the recycling inventory is prepared according to the advanced hydrometallurgical recycling process [25,43,45]. Additionally, both uses in mobility and stationary applications are calculated according to equations that estimate the electricity losses that each process has, and depend on variables such as battery mass, efficiency, and depth of discharge [30]. For the disposal process, a developed inventory was prepared here based on the production process by excluding the

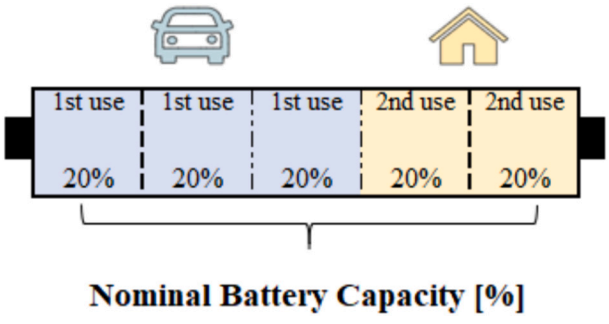


Fig. 2. Percentage distribution for attributing the emissions in the case of batteries also performing a second stationary application.

requirements of electricity, heat, and transport.

2.3. Life cycle impact assessment (LCIA)

The LCIA is performed using the open-source package *Brightway* to evaluate the contributions of each of the processes that the battery life cycle encompasses [51]. Besides, the Environmental Footprint impact (EF V3.1) assessment method is used to estimate 16 impact categories, which include, but are not limited to, acidification, climate change, human toxicity, eutrophication, water and land use, among others [12,52]. Additionally, we further include a prospective analysis, which is ideal for emerging technologies or technologies with an early phase of development, in order to represent their future performance with more maturity and nourish the LCA discussion [53,54]. Here, we use *Premise*, which is a tool developed to perform the prospective environmental impacts assessment of future energy systems (i.e. *p*-LCA), and helps to compare different energy policies [55]. Thanks to the inclusion of several Integrated Assessment Models (IAM), future environmental burdens are estimated by considering the interactions of the energy policies trajectories, emerging technologies, as well as market share, and efficiency of several technologies as part of the human development process [55]. IAM use Shared Socioeconomic Pathways (SSPs) and Representative Concentration Pathways (RCPs) to produce scenarios of how socioeconomic factors can influence future greenhouse gas emissions, respectively [56]. Here, IMAGE and REMIND are the IAM chosen to estimate the environmental emissions under SSP2 – "Middle of the road" narrative, where our global society follows a path in which social, economic, climate, and technological trends do not shift markedly from historical patterns, so they limit the radiative forcing to lower than 1.9 W/m^2, described as RCP19 or PkBudg500, respectively [56]. In conclusion, IMAGE (SSP2–RCP19) and REMIND (SSP2–PkBudg500) estimate the future environmental emissions by limiting the global mean surface temperature between 1.2 and 1.4 °C rise, in accordance with Paris Agreement objectives [57,58]. Furthermore, an ideal battery technical performance was not considered to quantify future potential climate change emissions, since there could be external limitations for the batteries to reach the foreseen optimal values of energy density, charge-discharge lifecycles, and round-trip efficiency, among others.

2.3.1. Prospective IAM scenarios: REMIND and IMAGE

Both REMIND and IMAGE are IAM, each with distinct focuses and methodologies, in order to understand and address complex environmental and socio-economic challenges [59,60]. REMIND emphasizes cost-efficient climate transformation pathways across sectors and regions [59], while IMAGE focuses on the broader interactions between human development and the natural environment, encompassing a wider range of environmental issues and policy implications [60].

REMIND (REgional Model of Investment and Development) is designed to investigate cost-efficient transformation pathways towards global or regional climate targets under various scenario assumption [59]. It comprises three main parts: a linear energy supply system, sector-specific energy demand representations, and a macroeconomic growth model [59,61]. These systems are interconnected through a non-linear intertemporal optimization of economic welfare, providing a high degree of endogeneity across all sectors. The model operates in a 21-world region setup, including detailed regional representations for the European Union and Germany [61]. Besides, it covers all relevant greenhouse gas emitting sectors and carbon dioxide removal options, featuring a comprehensive representation of the energy supply system (extraction, conversion, provision of energy) and detailed sector-specific modelling for energy demand (buildings, industry, transport). Land-use, agricultural emissions, bioenergy supply, and other land-based mitigation options are represented via reduced-form emulators. Furthermore, the scenario generation in REMIND is a two-stage process: first, input data are calibrated to GDP, population, and energy service trajectories for a reference scenario. Second, a climate policy scenario is run from

2025 to meet specific climate targets by adjusting a carbon price trajectory [61].

On the other hand, IMAGE (Integrated Model to Assess the Global Environment) is designed to address global environmental issues and sustainability challenges, such as climate change, land-use change, biodiversity loss, modified nutrient cycles, and water scarcity [60]. IMAGE relies on intermediate complexity modelling, balancing detail to capture critical processes with simplicity for multiple runs and sensitivity/uncertainty exploration [62]. While strong in representing the physical world and resource/technology selection, IMAGE has limitations. Its economy is represented across different model components, making it better suited for long-term trends than detailed economic impacts [62]. The model is aggregated for global coverage and a long-time horizon, meaning detailed local-scale processes and national policies are represented as part of global trends. It is well-adapted for studying technical measures but less so for specific policies like R&D policies, though some, like carbon taxes, can be represented [62]. Furthermore, IMAGE is used to explore two types of scenarios: how the future unfolds without deliberate changes in economic, technology, and policy developments (baseline scenario), and how policies and measures can prevent environmental impacts [62].

3. Results and discussion

The results obtained after applying the LCA methodology are analysed according to each approach studied, namely C2G or C2Gr, respectively. Additionally, the *p*-LCA analysis is shown for both approaches to see how future scenarios towards net-zero shape the environmental impacts of various battery technologies, depending on the type of application. Here, we mainly focus on the global warming potential (GWP) as an impact category for various battery technologies, yet other impacts can be found in the SI.

3.1. LCA analysis for battery cradle-to-gate approach

The Cradle-to-Gate (C2G) approach includes environmental burdens that are generated from raw material extraction, transporting of the raw materials to the factory, processing of the raw materials, and production and assembly of the batteries. Greenhouse gas emissions attributed to each battery component are shown in Fig. 3. Among the technologies analysed, NaNMMT and NaPBA generate the lowest GHG emissions with 51.5 and 66.5 kg CO2-eq/kWh, respectively, thanks to the exclusion of intensive raw materials such as lithium and cobalt, which are also associated with geographical limitations, since there are local constraints such as ore quality and technology for extraction [18,63]. For all the chemistries analysed, the cathode components are the main contributors (43–66%), followed by the anode (10–20%), except for the SSB, where the anode components (71%) are the largest contributor, followed by the cathode (22%). Here, 100% lithium metal is used in the anode for the SSB formulation, a critical raw material associated with emisions of 84.4 kg CO2-eq/kg of GHG, which are almost ten times higher than the emissions associated to graphite, 8.3 kg CO2-eq/kg [17]. These results show the latent need for batteries to include more earth crust's abundant components, such as sodium, iron, and magnesium, for the cathode formulation [64]. Besides, as anode active material of SIB, hard carbon is commonly produced by biomass generated from agricultural wastes, which generates less emissions than the graphite used in LIB and SSB [65,66].

Additionally, electricity and heat consumed along the battery value chains together contribute to 5–25% of the GHG emissions, which can be reduced by including more renewable energy technologies in the production process. The lowest contributions come from electrolyte and separator components (1.7%), in particular for SSB, which do not use solvents nor a plastic separator, thanks to their solid-state formulation. When comparing our results with previous publications under the C2G approach, our results are similar for NaMMMT, NaMVP, and NaPBA to those previously published [25,37], with less than 8% of difference. Discrepancies come mostly from the assumptions made for the share of electricity and heat for the production process, as well as the type of raw material used for the anode and cathode, namely, agricultural wastes, and the location of the extraction, respectively [25,37]. Besides, similar results for the GWP of NMC811 and NMC900 were published by Crenna et al. (2021) and Degen et al. (2024) with less than 5% of difference, mainly associated with the electricity and heat source. These authors

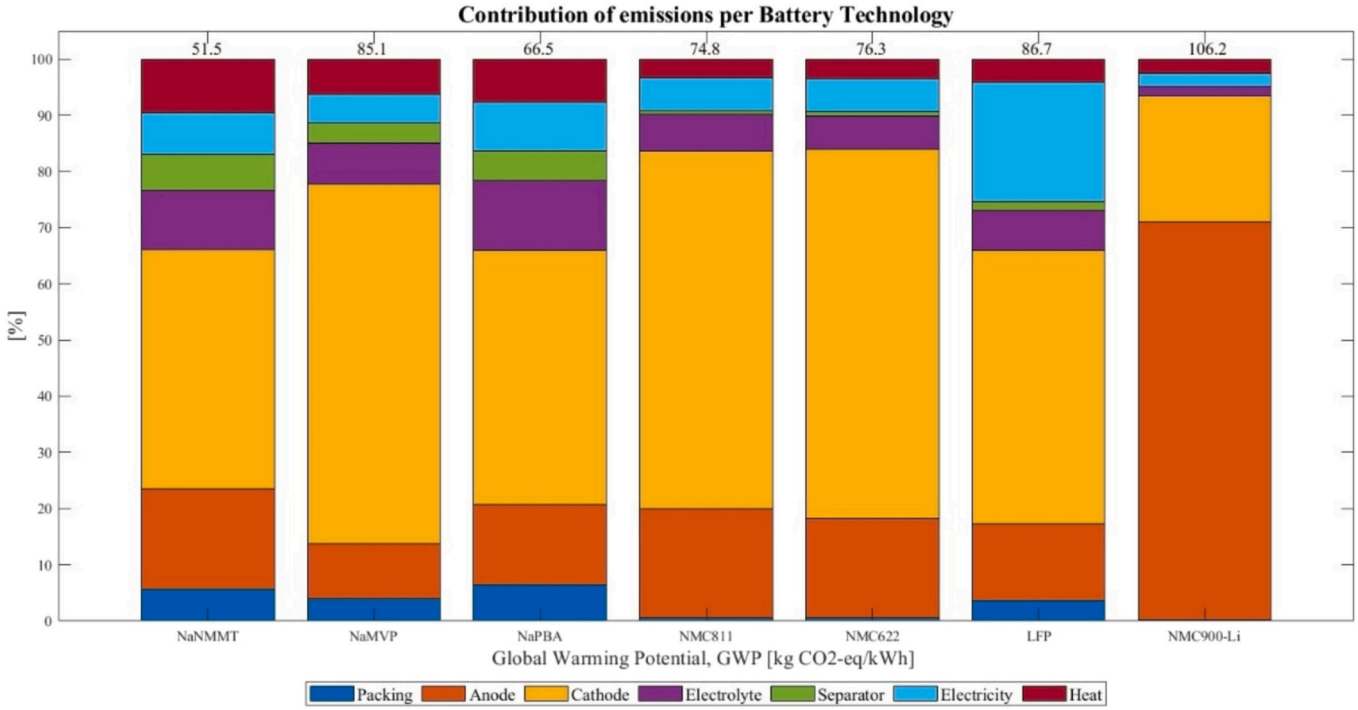


Fig. 3. Contributors of the GHG emissions per battery technology based on a functional unit of 1 kWh of storage capacity, cradle-to-gate approach.

used data from the previous literature and their own industrial process, respectively [17,38]. Additionally, our results for NMC622 and LFP also vary by 16% and 20%, respectively, with the previous literature, in particular due to small differences in the solvent and the electricity required for the cathode production [17].

3.2. *p*-LCA analysis for battery cradle-to-gate approach

Prospective scenarios are considered to estimate future GHG emissions throughout three different years, namely 2025 (base), 2030 and 2050, for both IMAGE (SSP2–RCP19) and REMIND (SSP2–PkBudg500) models, and compared with a baseline scenario (referred to as "base" during this study), as shown in Fig. 4.A, which corresponds to the results in Fig. 3. These IAM models consider potential transformations in sectors such as electricity, fuel generation, improvements in production and process efficiency to predict the environmental emissions that we can expect from these battery technologies in the future [37,57,58]. Our results based on scenarios IMAGE (SSP2–RCP19) and REMIND (SSP2–PkBudg500) show that the expected GHG reduction potential from SIB, LIB, and SSB ranges between 37–57%, 28–53% and 35–49%, between 2030 and 2050, respectively. Regardless of the model used for *p*-LCA, SIB offer a lower future GWP than LIB and SSB under a C2G approach. Prospective scenarios assume that emerging technologies with low battery component readiness level (BC-RL), such as SIB, tend to improve in energy efficiency to scale up to industrial manufacturing, and also reach wider market incorporation throughout the energy transition, in order to contribute to the future economic development, which directly reduces their emissions as resources are optimally used. Meanwhile, LIB is already a more mature technology, with a higher BC-RL, and thus, there is less room for improvement. In the case of LIB,

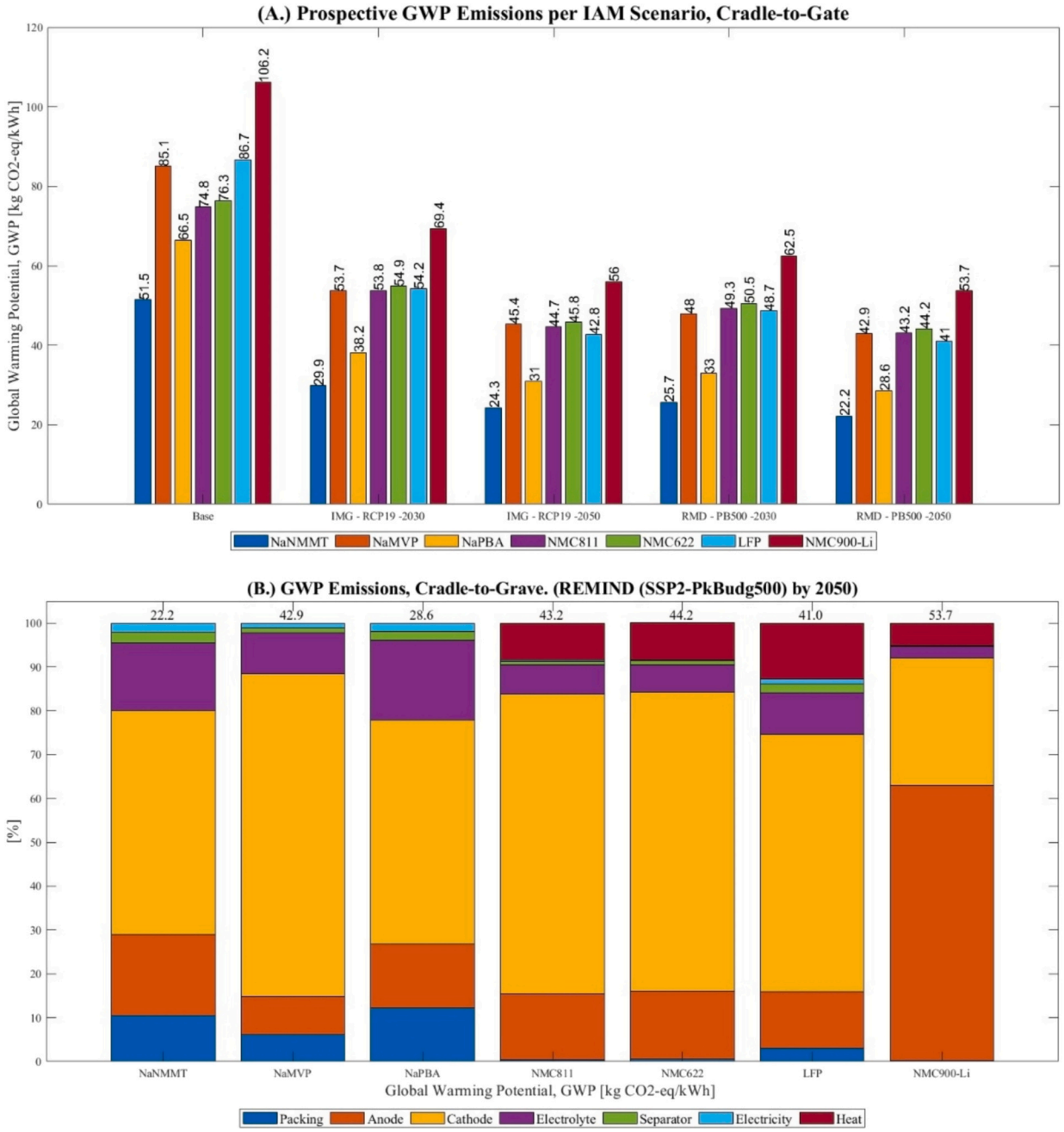


Fig. 4. Prospective GHG emissions under C2G approach: (A.) emissions expected per IAM scenario, (B.) emissions expected per battery technology under REMIND (SSP2-PkBudg500) by 2050.

emissions solely depend on the raw materials availability for production and the rise of the recycling process efficiency, except for LFP chemistry, which can be still optimised for future market implementation [67].

The REMIND (SSP2–PkBudg500) scenario predicts the lowest environmental emissions per technology, as shown in Fig. 4.B. Similarly to the battery cell production breakdown shown in Fig. 3 for the base case, the cathode components (51–74%) are the highest contributor of environmental emissions by 2050 regardless of the battery technology, followed by either the anode or electrolyte components, both between 9 and 18%. However, this is not the case for the NMC900, for which the anode (63%) is the predominant contributor of the environmental emissions since it contains lithium metal as active material, and it is followed by the cathode components (29%). As part of the REMIND scenario, the emissions related to electricity generation are reduced greatly, from 325.5 to 16.6 g CO2-eq/kWh, thanks to a renewable energy share of almost 95%. Subsequently, the electricity and heat contributions to the environmental emissions are reduced, when compared to the baseline scenario, around 95% for SIB, 47% for SSB, and 25% for LIB, and reductions from electrolyte and separators are 60%, 38% and 14% for SIB, LIB, and SSB, respectively.

3.3. LCA analysis for battery cradle-to-grave approach

The Cradle-to-Grave (C2Gr) approach encompasses not only considerations from C2G for the battery production, but also the application, namely mobility and/or stationary, as well as recycling and disposal. Here, we only focus on the batteries that performed better in the previous C2G approach analysed, as shown in Fig. 5. Thus, NaNMMT, NMC622, and NMC900 are only considered for the C2Gr analysis of SIB, LIB, and SSB, respectively, depending on the application scenario shown in Table 2. Results for all other environmental burdens under the C2Gr approach can be found in the SI.

3.3.1. Case study I

For this first case study, the environmental burdens under the C2Gr scope are related to the battery production and use in mobility applications, and their final disposal, for all battery technologies. Under current conditions, the GHG emissions generated show a large variation, mainly depending on the battery technology, and are 136.2, 183.8, and 240.1 kg CO2-eq/kWh for NaNMMT, NMC622, and NMC900, respectively, as shown in Fig. 5 – Case Study I. Besides, while incorporating a higher production and manufacturing process efficiency, based on the scenario improvements, as well as future renewable sources, the REMIND (SSP2–PkBudg500) scenario estimates a potential reduction of 66% for NaNMMT, 53% for NMC622, and 58% for NCM900 by the year 2050. Regardless of the year and scenario, the production and disposal processes are the highest contributors to GHG emissions for the studied batteries under case study I. Eventually, the environmental emissions attributed to the production process, and more specifically to the cathode active material, will be reduced thanks to the increasing percentage of recovered components that a new battery may possess [7]. The direct disposal of batteries is not a common practice since, preferably, critical raw materials are recovered and reused. However, this may be useful to illustrate other regions worldwide where there are no regulations on this matter yet. Currently, the EU is planning to increase the collection rate of batteries in the market to 100% by regulating that battery producers calculate the collection rate based on their annual battery sales [7].

3.3.2. Case study II

Under the C2Gr scope for case study II, the environmental burdens under the C2Gr scope are related to the battery production and use in mobility applications, before accounting for recycling (75%) and final disposal (25%). Under current conditions, the GHG emissions generated are 100.2, 93.6, and 101.7 kg CO2-eq/kWh for NaNMMT, NMC622, and NMC900, respectively, as shown in Fig. 5 – Case Study II. While incorporating a higher production and manufacturing process efficiency, as well as future renewable sources, the REMIND (SSP2–PkBudg500) scenario estimates a potential reduction of 74% for NaNMMT, and 68% for both NMC622 and NCM900 by the year 2050. Here, it can be easily observed that SIB do not have many benefits associated with their recycling process since current methods only target critical raw materials such as lithium, cobalt, nickel, and SIB incorporate a few of them

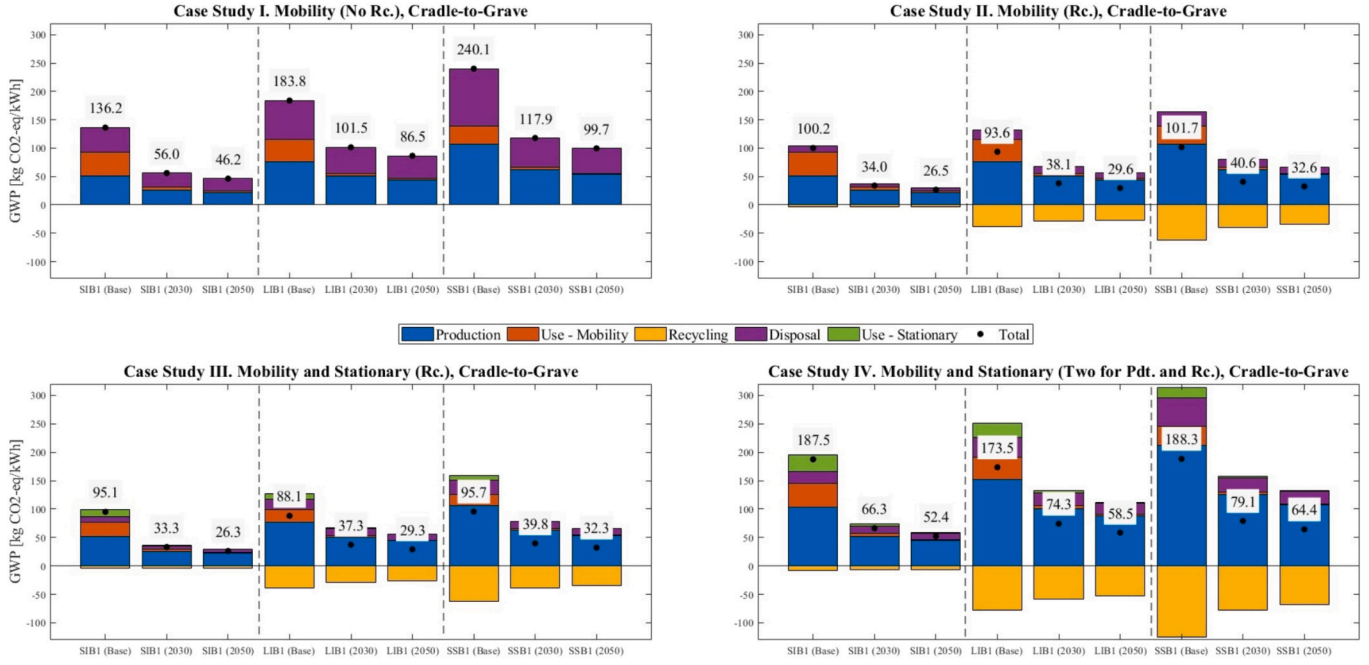


Fig. 5. GWP for SIB, LIB and SSB batteries according to each case study under C2Gr approach according to the REMIND (SSP2–PkBudg500) scenario: (**I**) the battery is used in mobility without recycling (No Rc.), (**II**) the battery is used in mobility with recycling (Rc.), (**III**) the battery is used first in mobility, followed by a stationary application with recycling (Rc.), and (**IV**) two different batteries are used, one for mobility, and another one for the stationary application, with recycling (Two for Pdt. and Rc.). Here, SIB1, LIB1 and SSB1 correspond to NaNMMT, NMC622 and NMC900-Li, respectively.

[25,43]. Therefore, improvements in the recovery of SIB raw materials should be addressed to compete with the other technologies, as well as the improvement of the recycling process efficiency, to reach the future market requirements [7,25]. Regardless of of the year and scenario, the production process is the highest contributor to GHG emissions for the studied batteries under case study II. Eventually, the environmental emissions attributed to the production process, and more specifically to the cathode active material, will be reduced thanks to the progressive incorporation of recovered components that new batteries may possess [7].

3.3.3. Case study III

For this case study, the environmental burdens under the C2Gr scope are generated by the production process, a first use in mobility (60%), a second use in stationary (40%), and finally, the recycling (75%) process and disposal (25%). Under current conditions, the GHG emissions generated by NaNMMT, NMC622, and NMC900 are 95.1, 88.1, and 95.7 kg CO2-eq/kWh, respectively, as shown in Fig. 5 – Case Study III. The potential environmental emissions reduction expected in the year 2050 by REMIND (SSP2–PkBudg500) is 72% for NaNMMT, 67% for NMC622, 66% for NMC900, and thanks to the improvements in the production process related to recovered raw materials and renewable supply assumed by 2050 [58]. This case study is the ideal life cycle that a battery can have to maximise the production and use of the natural resources. Yet, by working on increasing the recovered materials in the production process, extending the useful life during the use phases, and increasing the recovery rate of the recycling process, future environmental reductions can be further reduced [7,30].

3.3.4. Case study IV

For this case study, the environmental burdens under the C2Gr scope are generated by the production process of two different batteries (i.e. an overall use which corresponds to 200% taking *Case Studies I* and *II* as a reference): one battery is fully used in mobility (100%), and the second is also fully used in the stationary application (100%). It is assumed that recycling occurs for 75% of the overall battery capacity produced, while 25% is finally disposed. Under the current conditions, the GHG emissions generated by NaNMMT, NMC622, and NMC900 are 187.5, 173.5, and 188.3 kg CO2-eq/kWh, respectively, as shown in Fig. 5 – Case Study IV. The potential environmental emissions reduction expected in the year 2050, considering the REMIND (SSP2–PkBudg500) scenario, is 72% for NaNMMT, and 66% for both NMC622 and NMC900, thanks to the improvements in the production process related to recovered raw materials and renewable supply by 2050 [58]. We acknowledge that the capacity of the two batteries is not fully leveraged before they are recycled or disposed.

3.3.5. Case study comparison

First, the *Case Studies I* and *II* are compared to enlighten the benefits that recycling could bring, after a mobility application. Under current market technology conditions, the potential reductions attributed to including a recycling process after mobility applications are 36, 90.2, and 138.4 kg CO2-eq/kWh for NaNMMT, NMC622, and NMC900, respectively. Thus, the recycling process of critical raw materials reduces the GHG emissions in the range of 26–58%, depending on the battery technology. Additionally, the recycled materials, e.g., lithium and cobalt for LIB, and sodium and nickel for SIB, will be integrated as recovered materials into the production process of new batteries, thereby avoiding the extraction of fresh materials from ore deposits. New government regulations and directives should aim to increase both, high efficiency of recycling processes and the high content of recovered materials in new batteries [7].

On the other hand, the reduction of climate change emissions attributed to the avoidance of producing a new battery for the stationary application would be the difference between *Case Study III* and IV. Thus, having a combined application for the same battery could avoid 92.4, 85.4, and 92.6 kg CO2-eq/kWh for NaNMMT, NMC622, and NMC900, respectively. Thus, extending the battery lifecycle by having a secondary application can potentially reduce nearly 49% of GHG emissions. New government regulations and directives should aim to maximise the useful life of batteries by refurbishing them from one main application to another secondary, certifying battery safety and insurance, which needs to be further developed and included by policymakers [7,41,49,68].

By performing a sensitivity analysis on the recycling percentage that is applied, it was found that even with 100% of the battery recycling contributions, the studied battery technologies cannot reach a net-zero emissions goal by 2050, under the assumptions and input data considered in this study. Therefore, further actions are required to achieve the net-zero goal, beyond maximising the battery use. For instance, future policies targeting batteries should focus on R&D aiming to increase the energy density, which would enable storing and providing higher amounts of electricity when needed. Additionally, reducing ageing and increasing round-trip efficiency would also have a positive impact on climate change emissions. Furthermore, the recycling efficiency of other components that are not considered yet should be further improved, e.g., iron and magnesium, among others.

Next, we discuss the attribution of emisions for the two different applications, namely mobility and stationary. Here, the environmental emissions for the use phase of the battery are attributed as 60% and 40% for mobility and stationary applications, respectively (see Section 2.1.6 Case studies for details). In this study, the attribution of emissions is based on the total battery capacity loss, yet there are other options for this attribution. Thus, a sensitivity analysis on the percentage attribution of the use phase was performed to establish its real influence over the whole battery lifecycle considering our baseline scenario of emissions from 60% (M) and 40% (S), as shown in Fig. 6. It is found that by increasing the contribution of the second-life stationary application to 60% while reducing the share of the mobility application to 40% (40% (M) + 60% (S)), GHG emissions can be reduced by up to 3% for all the battery chemistries. In contrast, when the contribution of the first use in mobility is raised to 80% while the share of the second stationary application is decreased to 20% (80% (M) + 20% (S)), GWP emissions are increased by up to 3%. These results pinpoint the need to optimise the technology for the mobility application, even if a second application is expected.

3.3.6. Beyond climate change emissions

Studying other indicators beyond GHG emissions, such as acidification, ozone depletion, and water use, can be important to understand environmental trade-offs for various battery technologies, as shown in Fig. 7 [12,52]. Here, we have chosen the indicators mainly used in the previous literature to illustrate the baseline and future environmental emissions, with the latter based on the REMIND (SSP2–PkBudg500) scenario for SIB, LIB, and SSB. Here, only the case study III, which includes both a second mobility application and recycling is considered, since it is the best in terms of GHG emissions.

In terms of acidification, SIB generate less emissions to the air, thus reducing the deposition of emissions in water and soil, in comparison to LIB and SSB, up to the year 2050. Moreover, the production phase is the main contributor to acidification, regardless of the battery technology, since acidification is related to the decline of forests and increase in fish mortality due to the sulphur content attributed to mining extraction, electricity generation, heat production, and transport, in particular, contributing between 62-80% [25,52]. Thus, the raw material extraction for LIB and SSB contributes a higher proportion to the accumulated exceedance [mol H+-eq/kWh], related to copper, nickel, and cobalt during the roasting of these sulfidic ores [25]. Besides, the recycling process reduces the emissions that cause acidification for all the studied batteries. For instance, the benefits to SIB during recycling come mainly from the hard carbon recuperation. Further, for LIB and SSB, the recovery of the mentioned raw materials could avoid the release of extra emissions to air, water, and soil. Similar behaviour is observed with the

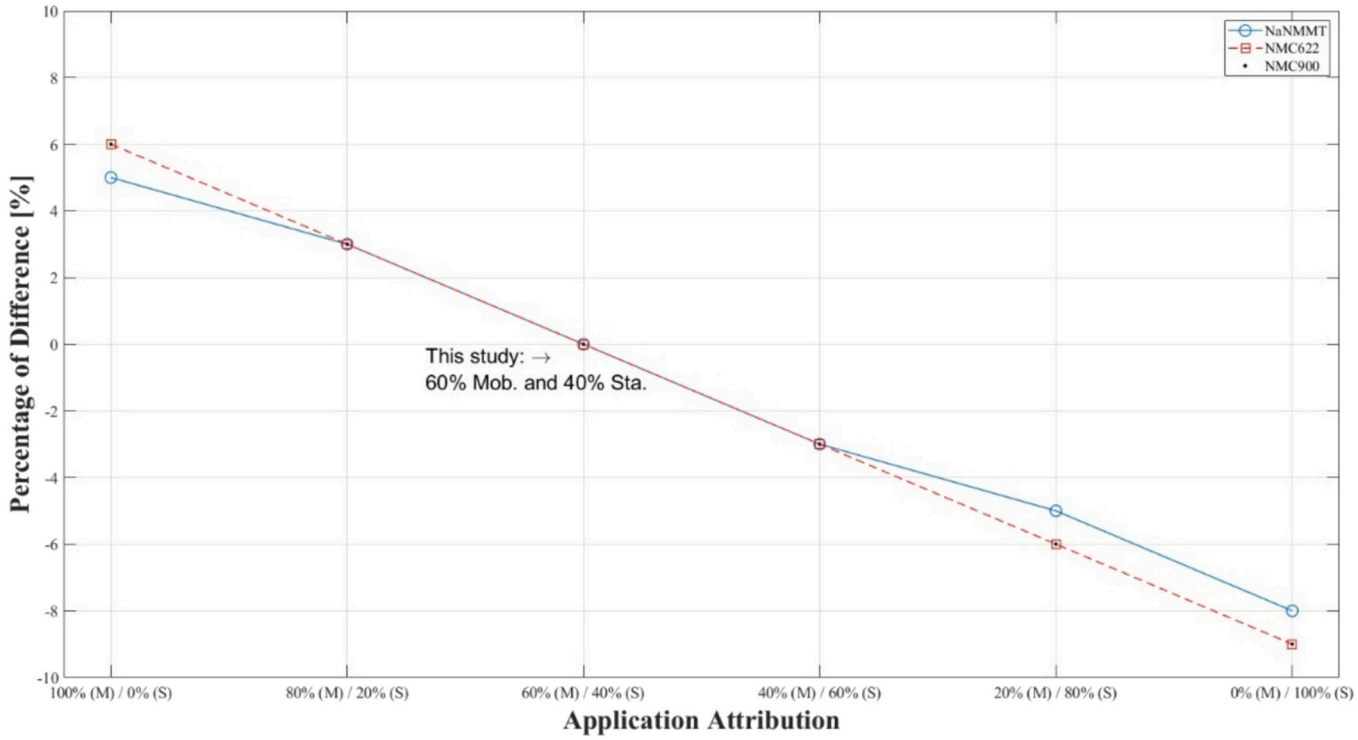


Fig. 6. Sensitivity analysis for the attribution of the emissions, case study III. Where M refers to mobility, and S to stationary.

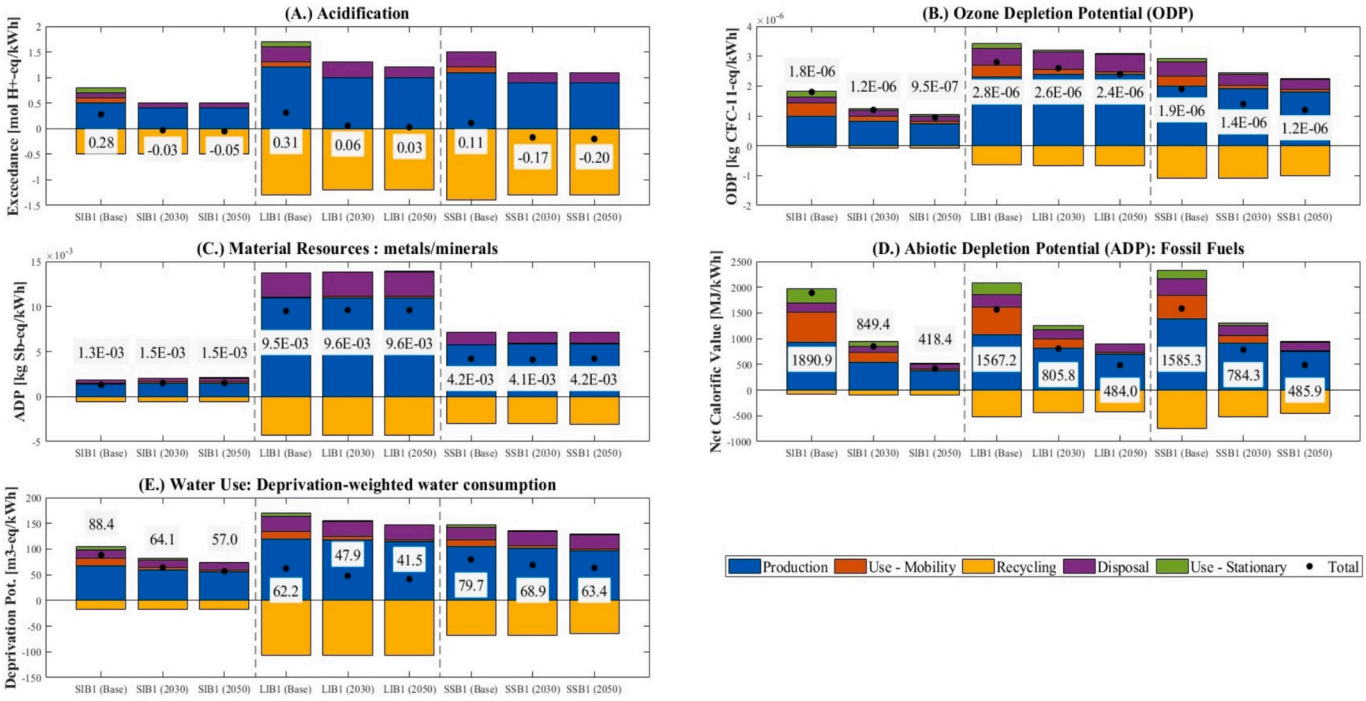


Fig. 7. Other category indicators for SIB, LIB, and SSB batteries under the C2Gr approach according to the REMIND (SSP2–PkBudg500) scenario for case study III: (A.) Acidification, (B.) Ozone depletion potential, (C.) Material resources, (D.) Abiotic Depletion Potential (ADP), (E.) Water use. Here, SIB1, LIB1 and SSB1 correspond to NaNMMT, NMC622 and NMC900-Li, respectively.

indicator ozone depletion potential, since SIB generate fewer substances to deteriorate the stratospheric ozone layer that protects us from hazardous ultraviolet radiation (UV-B), which is also associated with potential damage to humans, animals, and plants [52]. Here again, the production process contributes 54–80% of the ODP-related emissions of the studied battery technologies, which are mainly related to the need for specific temperature and humidity conditions required during the process. Therefore, the recycling benefits are not as predominant as for the previous indicator.

The material resources indicator is based on the use of minerals and metals that are currently available on earth, so a high extraction of a component today will require future generations to extract lower

concentration or lower value resources [52]. Thus, technologies such as LIB (9.6 g Sb-eq/kWh) and SSB (4.2 g Sb-eq/kWh) that require critical raw materials will have 640% and 280% higher depletion potential when compared to formulations that incorporate materials that are more abundant in the earth's crust, namely SIB (1.5 g Sb-eq/kWh) [25]. Thereby, there should be a balance to incorporate both critical and abundant raw materials without compromising the expected technical performance of the battery. Additionally, the abiotic depletion potential (ADP) of fossil fuels quantifies the use of non-renewable resources such as coal, oil, and gas [52]. Similarly, the use of finite resources will leave future generations with fewer resources and in search of new ones. In terms of ADP, benefits from the recycling process enable LIB and SSB to generate lower emissions than SIB in the current market conditions, mainly based on the recovery of critical raw materials [25]. However, under the scope of the REMIND scenario by 2050, SIB could potentially reduce the energy consumption in the production process as industrial scaling and mass production optimisation take place.

Furthermore, the water use indicator based on user deprivation potential shows the abstraction of available water from lakes, rivers, or groundwater. This impact category could consider the availability or scarcity of water in the regions where the activity takes place, if this information is available [52]. Since material extraction requires large amounts of water during mining activities, LIB and SSB show higher user deprivation potential, which could be compensated for by the recovery of critical raw materials that will not be taken from the soil. However, independently of the battery technology, all the studied battery technologies will require water for both production and recycling processes, and wastewater treatment plants are highly recommended to retrieve and reuse this limited natural resource and mitigate any potential water stress.

Based on our results, we show that decision-makers should not only consider GHG emissions to select the optimal battery technology for mobility, including a second stationary application. Interestingly, batteries that generate high GWP emissions, such as LIB and SSB, also cause high acidification, ozone depletion, material resource, and water use impacts. In contrast, SIB generate low GHG emissions as well as offering lower environmental impacts for the other indicators mentioned above. Therefore, when considering all the indicators quantified in this study, we could expect less environmental burden for technologies that do not incorporate critical raw materials, namely SIB.

4. Conclusions

This paper analyses and compares the environmental burdens of LIB, SIB, and SSB depending on the type of application, namely mobility and mobility combined with a second stationary application, following the LCA methodology. We quantify GHG emissions and other environmental burdens related to LIB, SIB, and SSB technologies both under the current energy and technology market conditions. Additionally, by considering future improvements in energy source, production, and process efficiencies, which are modelled with *p*-LCA, in particular the REMIND (SSP2–PkBudg500) and IMAGE (SSP2–RCP19) integrated assessment models for the proposed battery technologies, we determined various relevant future indicators.

It was found that under a C2G approach, LIB, SIB, and SSB should increase the incorporation of recovered raw materials to reduce GWP attributed to virgin raw materials. Regardless of the application, when a C2Gr approach is considered, the recycling process can increase the overall material efficiency by recovering other raw materials that are not yet considered, such as manganese, magnesium, phosphorus, and iron. Despite the battery manufacturing, use and recycling remain the key hotspots for environmental impacts. Thus, we recommend that policy-makers and technology providers request similar battery arrangements following eco-design principles to easily assemble and disassemble to promote a second stationary application. With a proper battery eco-design, the disposable materials for repurposing the battery from the first to the second application will be reduced. Furthermore, based on the results obtained from the prospective analysis considering the different impact category indicators analysed, SIB will offer lower environmental burdens by 2050, while further work should be accomplished with LIB and SSB in terms of their production process to reduce their future environmental emissions. Importantly, a balance between critical and abundant raw materials in the battery formulation is encouraged, both to reach high technical performance and long-term environmental sustainability. Thus, regardless of the technology, our results pinpoint the need for further improvements for all the studied battery technologies in terms of energy density, charge-discharge lifecycles, and round-trip efficiency to reach the net-zero emissions goal by 2050.

To conclude, this is the first paper to determine the environmental impacts of various battery technologies, including SIB and SSB, in addition to LIB, performing mobility and a second stationary application, including prospective LCA. Importantly, a sensitivity analysis is performed to understand the impact of the attribution on the final emissions for both applications, depending on the battery technology. Yet, in future work, this model could be reused and extended to represent the capacity decay per battery technology in more detail depending on the type of application. Additionally, future work should focus on the other pillars of the Life Cycle Sustainable Assessment (LCSA) for LIB, SIB, and SSB technologies, to further include the economic and social aspects that should be expected from emerging technologies. The economic aspect can be covered by different methodologies, such as the pseudo-two-dimensional (P2D) battery cell model and a bottom-up assessment framework. Another option is the commodity life cycle costing (C-LCC) approach, which establishes an indicator by considering costs related to natural resources used during the life cycle. The lack of research focusing on social-LCA is a constant in the studied batteries, and it should be considered to address the human and social issues that each battery technology could bring to the market and its industrial implementation.

CRediT authorship contribution statement

Luis Alfredo Carmona Arrieta: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Paula Sanz Camacho:** Writing – review & editing. **Juan M. Perez:** Writing – review & editing. **Christian Bauer:** Writing – review & editing, Supervision, Methodology, Conceptualization. **David Parra:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was possible thanks to funding received through the European Union's Next Generation EU funds, articulated at the national level by the Recovery, Transformation and Resilience Plan. Also, this work has been funded by the Ministry of Science, Innovation and Universities, and the Regional Government of Extremadura through the CIIAE Unit.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.est.2026.121839.

Data availability

Supplementary information has been provided as an attachement.

References

[1] United Nations, "Paris Agreement, Volume 3156, I - 54113. Registration with the Secretariat of the United Nations: ex officio, 4 November 2016," Paris, Accessed: Oct. 16, 2025. [Online]. Available: https://treaties.un.org/doc/Publication/UNTS/Volume%203156/Part/volume-3156-I-54113.pdf, Dec. 2015.

[2] European Commission, A strategic long-term vision for a prosperous, modern, competitive and climate-neutral EU economy climate action, Accessed: Jun. 04, 2024. [Online]. Available: https://data.europa.eu/doi/10.2834/02074, 2019.

[3] IRENA, Global renewables outlook energy transformation 2050, Accessed: Jun. 06, 2024. [Online]. Available:, Abu Dhabi, 2020. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2020/Apr/IRENA_Global_Renewables_Outlook_2020.pdf?rev=1f416406e50d447cbb2247de30d1d1d0.

[4] IRENA, Tripling renewable power by 2030: the role of the G7 in turning targets into action, Accessed: Jun. 06, 2024. [Online]. Available:, Abu Dhabi, 2024. https://mc-cd8320d4-36a1-40ac-83cc-3389-cdn-endpoint.azureedge.net/-/media/Files/IRENA/Agency/Publication/2024/Apr/IRENA_G7_Tripling_renewable_power_2024.pdf?rev=4d47c35c8d824fbd8bd6dfb28cede677.

[5] IRENA, 100% renewable energy scenarios, Accessed: Jun. 06, 2024. [Online]. Available:, Abu Dhabi, 2024. https://mc-cd8320d4-36a1-40ac-83cc-3389-cdn-endpoint.azureedge.net/-/media/Files/IRENA/Agency/Publication/2024/Mar/IRENA_Coalition_100_RE_scenarios_2024.pdf?rev=5ef440b6087942e8a8eba687b65c25f9.

[6] IEA, "World Energy Outlook Special Report Batteries and Secure Energy Transitions," Paris, Accessed: Jun. 06, 2024. [Online]. Available: https://iea.blob.core.windows.net/assets/cb39c1bf-d2b3-446d-8c35-aae6b1f3a4a0/BatteriesandSecureEnergyTransitions.pdf, Apr. 2024.

[7] Regulation (EU) 2023/1542, Regulation (EU) 2023/1542 of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC [2023] OJ L 191/1, Accessed: Jun. 04, 2025. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1542, Jul. 2023.

[8] ISO 14044, "ISO 14044: environmental management — life cycle assessment — requirements and guidelines.," Geneva, Switzerland, Accessed: Oct. 16, 2025. [Online]. Available: https://www.iso.org/standard/38498.html#lifecycle, 2006.

[9] ISO 14040, "ISO 14040: environmental management — life cycle assessment — principles and framework.," Geneva, Switzerland, Accessed: Oct. 16, 2025. [Online]. Available: https://www.iso.org/standard/37456.html#amendment, 2006.

[10] ISO 14072, "ISO 14072: environmental management — life cycle assessment — requirements and guidelines for organizational life cycle assessment.," Geneva, Switzerland, Accessed: Oct. 16, 2025. [Online]. Available: https://www.iso.org/standard/86265.html, 2024.

[11] J.B. Guinée, et al., Life cycle assessment an operational guide to the ISO standards, Accessed: Jul. 25, 2024. [Online]. Available: https://link.springer.com/book/10.1007/0-306-48055-7, 2001.

[12] Commission Recommendation (EU) 2021/2279, Commission Recommendation (EU) 2021/2279 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations [2021] OJ L 471/1, Accessed: Oct. 02, 2025. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021H2279, Dec. 2021.

[13] J.F. Peters, M. Baumann, B. Zimmermann, J. Braun, M. Weil, The environmental impact of Li-Ion batteries and the role of key parameters – a review, Renew. Sustain. Energy Rev. 67 (2017) 491–506, https://doi.org/10.1016/j.rser.2016.08.039.

[14] IEA, "World energy outlook special report strategies for affordable and fair clean energy transitions," Paris, Accessed: Jun. 06, 2024. [Online]. Available: https://iea.blob.core.windows.net/assets/59c6e72a-7cc4-403d-b521-7375114baca9/StrategiesforAffordableandFairCleanEnergyTransitions.pdf, May 2024.

[15] A. Temporelli, M.L. Carvalho, P. Girardi, Life cycle assessment of electric vehicle batteries: an overview of recent literature, Energies 13 (11) (2020) 2864, https://doi.org/10.3390/en13112864.

[16] X. Lai, et al., Critical review of life cycle assessment of lithium-ion batteries for electric vehicles: a lifespan perspective, eTransportation 12 (2022) 100169, https://doi.org/10.1016/j.etran.2022.100169.

[17] F. Degen, M. Mitterfellner, A. Kampker, Comparative life cycle assessment of lithium-ion, sodium-ion, and solid-state battery cells for electric vehicles, J. Ind. Ecol. 29 (1) (2024) 113–128, https://doi.org/10.1111/jiec.13594.

[18] H. Ren, et al., Vulnerability to geopolitical disruptions of the global electric vehicle lithium-ion battery supply chain network, Comput. Ind. Eng. 188 (Feb. 2024), https://doi.org/10.1016/j.cie.2024.109919.

[19] G. Majeau-Bettez, T.R. Hawkins, A.H. StrØmman, Life cycle environmental assessment of lithium-ion and nickel metal hydride batteries for plug-in hybrid and battery electric vehicles, Environ. Sci. Technol. 45 (10) (May 2011) 4548–4554, https://doi.org/10.1021/es103607c.

[20] Q. Dai, J.C. Kelly, L. Gaines, M. Wang, Life cycle analysis of lithium-ion batteries for automotive applications, Batteries 5 (2) (Jun. 2019), https://doi.org/10.3390/batteries5020048.

[21] Q. Chen, et al., A critical comparison of LCA calculation models for the power lithium-ion battery in electric vehicles during use-phase, Energy 296 (Jun. 2024) 131175, https://doi.org/10.1016/j.energy.2024.131175.

[22] M. Safari, A perspective on the battery value chain and the future of battery electric vehicles, Battery Energy 4 (1) (Jan. 2025), https://doi.org/10.1002/bte2.20240016.

[23] Q. Chen, et al., Environmental impacts and supply risks for LiFePO4 - LiCo Ni Mn O2 hybrid battery pack in China, Process Saf. Environ. Prot. 198 (Jun. 2025) 107115, https://doi.org/10.1016/j.psep.2025.107115.

[24] F. Rossi, M.L. Parisi, S. Greven, R. Basosi, A. Sinicropi, Life cycle assessment of classic and innovative batteries for solar home systems in Europe, Energies (Basel) 13 (13) (Jul. 2020), https://doi.org/10.3390/en13133454.

[25] J.F. Peters, M. Baumann, J.R. Binder, M. Weil, On the environmental competitiveness of sodium-ion batteries under a full life cycle perspective-a cell-chemistry specific modelling approach, 2021, https://doi.org/10.5281/zenodo.4742246.

[26] F.B. Jasper, J. Späthe, M. Baumann, J.F. Peters, J. Ruhland, M. Weil, Life cycle assessment (LCA) of a battery home storage system based on primary data, J. Clean. Prod. 366 (Sep. 2022), https://doi.org/10.1016/j.jclepro.2022.132899.

[27] T.S. Schmidt, et al., Additional emissions and cost from storing electricity in stationary battery systems, Environ. Sci. Technol. 53 (7) (Apr. 2019) 3379–3390, https://doi.org/10.1021/acs.est.8b05313.

[28] H. Kang, et al., Life-cycle environmental impacts of reused batteries of electric vehicles in buildings considering battery uncertainty, Renew. Sustain. Energy Rev. 207 (Jan. 2025), https://doi.org/10.1016/j.rser.2024.114936.

[29] T. Kobayashi, H. Kondo, T. Sasaki, Life cycle assessment integrating the effects of recycling and reuse for battery circulation, J. Power Sources 624 (Dec. 2024), https://doi.org/10.1016/j.jpowsour.2024.235544.

[30] W. Guo, T. Feng, W. Li, L. Hua, Z. Meng, K. Li, Comparative life cycle assessment of sodium-ion and lithium iron phosphate batteries in the context of carbon neutrality, J. Energy Storage 72 (Nov. 2023), https://doi.org/10.1016/j.est.2023.108589.

[31] M.G.M. Abdolrasol, S. Ansari, S.R. Arsad, T.S. Kiong, M.A. Hannan, Solid-state battery: An emerging transformation of battery technology in electric vehicle applications, J. Energy Storage 140 (Dec. 2025), https://doi.org/10.1016/j.est.2025.118988.

[32] P. Mandade, M. Weil, M. Baumann, Z. Wei, Environmental life cycle assessment of emerging solid-state batteries: a review, Chem. Eng. J. Adv. 13 (Mar. 2023), https://doi.org/10.1016/j.ceja.2022.100439.

[33] S. Zhang, D. Brandell, M. Valvo, B. Steubing, Å. Nordberg, Future climate impact of all-solid-state batteries, J. Clean. Prod. 525 (Sep. 2025), https://doi.org/10.1016/j.jclepro.2025.146607.

[34] A. Spindlegger, L. Slotyuk, A. Jandric, R.G. De Souza, S. Prenner, F. Part, Environmental performance of second-life lithium-ion batteries repurposed from electric vehicles for household storage systems, Sustain. Prod. Consum. 54 (Mar. 2025) 227–240, https://doi.org/10.1016/j.spc.2025.01.003.

[35] A. Mohanty, et al., Second-life batteries for a greener grid: strategies for sustainable energy storage and e-waste reduction, J. Energy Storage 143 (Jan. 2026) 119750, https://doi.org/10.1016/j.est.2025.119750.

[36] M.F. Roslan, P.R. Satpathy, T. Prasankumar, V.K. Ramachandaramurthy, M. Mansor, S.L. Walker, Second-life battery energy storage system for energy sustainability: recent advancements, key takeaways and future perspectives 123 (2025) 116808, https://doi.org/10.1016/j.est.2025.116808.

[37] S. Zhang, B. Steubing, H. Karlsson Potter, P.A. Hansson, Å. Nordberg, Future climate impacts of sodium-ion batteries, Resour. Conserv. Recycl. 202 (Jan. 2024), https://doi.org/10.1016/j.resconrec.2023.107362.

[38] E. Crenna, M. Gauch, R. Widmer, P. Wäger, R. Hischier, Towards more flexibility and transparency in life cycle inventories for Lithium-ion batteries, Resour. Conserv. Recycl. 170 (Jul. 2021), https://doi.org/10.1016/j.resconrec.2021.105619.

[39] M. Zackrisson, L. Avellán, J. Orlenius, Life cycle assessment of lithium-ion batteries for plug-in hybrid electric vehicles-critical issues, J. Clean. Prod. 18 (15) (2010) 1519–1529, https://doi.org/10.1016/j.jclepro.2010.06.004.

[40] H.C. Kim, T.J. Wallington, J.L. Sullivan, G.A. Keoleian, Life cycle assessment of vehicle lightweighting: novel mathematical methods to estimate use-phase fuel consumption, Environ. Sci. Technol. 49 (16) (Jul. 2015) 10209–10216, https://doi.org/10.1021/acs.est.5b01655.

[41] M. Philippot, et al., Environmental impact of the second life of an automotive battery: reuse and repurpose based on ageing tests, J. Clean. Prod. 366 (Sep. 2022), https://doi.org/10.1016/j.jclepro.2022.132872.

[42] W. Clarke, Car Warranty Coverage on an Electric Car Battery, U.S. News, Dec. 23, 2024.

[43] M. Mohr, J.F. Peters, M. Baumann, M. Weil, Toward a cell-chemistry specific life cycle assessment of lithium-ion battery recycling processes, J. Ind. Ecol. 24 (6) (Dec. 2020) 1310–1322, https://doi.org/10.1111/jiec.13021.

[44] K. Fisher, E. Wallén, P.P. Laenen, M. Collins, Delivering Sustainable Solutions in a More Competitive World Battery Waste Management Life Cycle Assessment Final Report for Publication Battery Waste Management Life Cycle Assessment Final Report for Publication, 2006.

[45] Duesenfeld GmbH, "Method for the utilization of lithium batteries," EP 3517641 B1, Accessed: Jun. 10, 2025. [Online]. Available: https://data.epo.org/publication-server/rest/v1.0/publication-dates/20200325/patents/EP3517641NWB1/document.pdf, 2020.

[46] X. Han, M. Ouyang, L. Lu, J. Li, A comparative study of commercial lithium ion battery cycle life in electric vehicle: capacity loss estimation, J. Power Sources 268 (Dec. 2014) 658–669, https://doi.org/10.1016/j.jpowsour.2014.06.111.

[47] J. Yang, F. Gu, J. Guo, Environmental feasibility of secondary use of electric vehicle lithium-ion batteries in communication base stations, Resour. Conserv. Recycl. 156 (May 2020), https://doi.org/10.1016/j.resconrec.2020.104713.
[48] D. Parra, S.A. Norman, G.S. Walker, M. Gillott, Optimum community energy storage for renewable energy and demand load management, Appl. Energy 200 (2017) 358–369, https://doi.org/10.1016/j.apenergy.2017.05.048.
[49] J. Wesselkämper, T.P. Hendrickson, S. Lux, S. von Delft, Recycling or second use? Supply potentials and climate effects of end-of-life electric vehicle batteries, Environ. Sci. Technol. 59 (30) (Aug. 2025) 15751–15765, https://doi.org/10.1021/acs.est.5c01823.
[50] R. Sacchi, PRospective EnvironMental Impact AsSEssment, in: Premise, data, additional inventories, 2020. Accessed: Jan. 01, 2025. [Online]. Available: https://github.com/polca/premise/tree/5af46bd3c78aef20805e9a9f704b226ff6341986/premise/data/additional_inventories.
[51] C. Mutel, Brightway: an open source framework for Life Cycle Assessment, J. Open Source Softw. 2 (12) (Apr. 2017) 236, https://doi.org/10.21105/joss.00236.
[52] European Commission, Life Cycle Assessment: the scientific foundation of the Environmental Footprint methods, Accessed: Oct. 02, 2025. [Online]. Available: https://green-forum.ec.europa.eu/green-business/environmental-footprint-methods/life-cycle-assessment-ef-methods_en, 2021.
[53] C. van der Giesen, S. Cucurachi, J. Guinée, G.J. Kramer, A. Tukker, A critical view on the current application of LCA for new technologies and recommendations for improved practice, J. Clean. Prod. 259 (Jun. 2020), https://doi.org/10.1016/j.jclepro.2020.120904.
[54] S. Cucurachi, C. Van Der Giesen, J. Guinée, Ex-ante LCA of Emerging Technologies, in: Procedia CIRP, Elsevier B.V., 2018, pp. 463–468, https://doi.org/10.1016/j.procir.2017.11.005.
[55] R. Sacchi, et al., PRospective EnvironMental Impact asSEment (premise): a streamlined approach to producing databases for prospective life cycle assessment using integrated assessment models, Renew. Sustain. Energy Rev. 160 (May 2022), https://doi.org/10.1016/j.rser.2022.112311.
[56] Z. Hausfather, Explainer: how 'Shared Socioeconomic Pathways' explore future climate change, in: Carbon Brief, Clear on Climate. Climate Modelling, Apr. 19, 2018.
[57] E. Stehfest, D. van Vuuren, T. Kram, L. Bouwman, Integrated Assessment of Global Environmental Change with IMAGE 3.0: Model description and policy applications, Accessed: Jun. 11, 2025. [Online]. Available: https://www.pbl.nl/uploads/default/downloads/pbl-2014-integrated_assessment_of_global_environmental_change_with_image30_735.pdf, 2014.
[58] L. Baumstark, et al., REMIND2.1: transformation and innovation dynamics of the energy-economic system within climate and sustainability limits, Geosci. Model Dev. 14 (10) (Oct. 2021) 6571–6603, https://doi.org/10.5194/gmd-14-6571-2021.
[59] T.I.A.M.C. IAMC, REMIND - REgional Model of Investment and Development, Accessed: Jan. 23, 2026. [Online]. Available:, January 7, 2021 https://www.iamconsortium.org/resources/model-resources/remind/.
[60] T.I.A.M.C. IAMC, IMAGE - Integrated Model to Assess the Global Environment, Accessed: Jan. 23, 2026. [Online]. Available:, August 10, 2020 https://www.iamconsortium.org/resources/model-resources/image/.
[61] K.P. Ariadne and German Federal Ministry of Research Technology and Space, Model Documentation: REMIND, 2021. Accessed: Jan. 23, 2026. [Online]. Available: https://ariadneprojekt.de/en/model-documentation-remind/.
[62] PBL - Netherlands Environmental Assessment Agency, IMAGE framework, 1 November 2022. Accessed: Jan. 23, 2026. [Online]. Available: https://models.pbl.nl/index.php?title=IMAGE_framework&oldid=36998.
[63] M. Grohol, C. Veeh, D.G. Grow, European Commission, Study on the Critical Raw Materials for the EU 2023 – Final Report, 2023, https://doi.org/10.2873/725585.
[64] M. Baumann, et al., Prospective sustainability screening of sodium-ion battery cathode materials, Adv. Energy Mater. 12 (46) (Dec. 2022), https://doi.org/10.1002/aenm.202202636.
[65] H. Liu, et al., A life cycle assessment of hard carbon anodes for sodium-ion batteries, Philos. Trans. R. Soc. A: Math. Phys. Eng. Sci. 379 (2209) (Nov. 2021), https://doi.org/10.1098/rsta.2020.0340.
[66] Z. Xu, et al., The role of hydrothermal carbonization in sustainable sodium-ion battery anodes, Adv. Energy Mater. 12 (18) (May 2022), https://doi.org/10.1002/aenm.202200208.
[67] M. Greenwood, J.M. Wrogemann, R. Schmuch, H. Jang, M. Winter, J. Leker, The Battery Component Readiness Level (BC-RL) framework: a technology-specific development framework 14 (2022) 100089, https://doi.org/10.1016/j.powera.2022.100089.
[68] B. Wrålsen, R. O'Born, Use of life cycle assessment to evaluate circular economy business models in the case of Li-ion battery remanufacturing, Int. J. Life Cycle Assess. 28 (5) (May 2023) 554–565, https://doi.org/10.1007/s11367-023-02154-0.